How-To Library

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# Performing Data Coverage Analysis in Test Data Management

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# Abstract

You can perform data coverage analysis on test data that you store in the test data warehouse in Test Data Management (TDM). This article provides the steps to perform data coverage analysis in TDM with a specific example. This article assumes that you have knowledge of TDM and are familiar with TDM operations.

# Supported Versions

- Test Data Management 10.2.0 HotFix 2
- Test Data Management 10.2.0 HotFix 1
- Test Data Management 10.2 R1.0
- Test Data Management 10.2.0
- Test Data Management 10.1.0

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# Overview

In TDM, you can analyze test data to view data density and to check whether you have a minimum amount of data for specific test cases.

Create a data coverage task to perform pairwise data analysis and to create a visual representation of the data coverage of the test data. You can assess the quality of test data by analyzing combinations of values in any two columns in the task. You can change the combinations to ensure that you cover all valid combinations of values. You can improve the quality of the data and move data across categories to meet the minimum data threshold that you require.

To perform data coverage analysis and create the required test data, perform the following steps:

1. Create a data coverage task.
2. Create columns to use in the task.
3. Use the columns to perform data coverage analysis.
4. Update data in columns based on results of the analysis and data requirements.

## Example

You work for a test team that uses TDM to create and manage test data in the test data warehouse. You are responsible for test data creation for specific test cases.

You create and store the test data in the test data warehouse. Before you tag a data set for a test case, you need to analyze the availability of required data and update the data if needed.

A set of test cases requires data on customer orders placed in every 10-day range of the month. To ensure you have data that meets this requirement, you must analyze the invoice date data and the customer order data. You create a data coverage task in Test Data Manager and create columns to analyze the data. Create a column that uses ranges for the invoice data and a column that uses the data as is for the customer orders data. The data must contain records for every 10-day range of the month. If a range does not contain data, update the data to ensure that the data set contains data on customer orders in all date ranges.

This document describes the steps to create a data coverage task and perform the analysis. The task includes a column based on ranges and a column that uses the data as is. Based on the results of the analysis, you update the data in required columns.

## Prerequisites

Before you perform data coverage analysis, perform the following tasks:

- Verify that the Test Data Management version supports the test data warehouse.
- Log in to Test Data Manager and create a data set that contains test data that you want to analyze.

## Rules and Guidelines

Read the following rules and guidelines before you perform data coverage analysis:

- Review the list of data type exceptions in this document before you create a data coverage column.
- You can use ranges to analyze data that is in numeric or date data type.
- Based on the schema, a move operation might fail in some cases.
- You cannot update data in a cell that contains data of the Blob data type.

## Data Type Exceptions for Data Coverage Columns

You cannot create data coverage columns with source data columns that contain certain data types.

The following table lists the data types that are unavailable when you create data coverage columns:

| Data Coverage Column Type | Data Type |
|---|---|
| Use As Is | You cannot use the following data types to create data coverage columns that use data as is:<br>- Date<br>- Real<br>- Double<br>- Precision<br>- Decimal<br>- Decimal (p,s)<br>- Float<br>- Binary_float<br>- Binary_double |
| Range | String |
| Mapping | You cannot use the following data types to create data coverage columns that use mappings:<br>- Date<br>- Real<br>- Double<br>- Precision<br>- Decimal<br>- Decimal (p,s)<br>- Float<br>- Binary_float<br>- Binary_double |

You cannot include the following data types in data coverage tasks:

- Dburitype
- Xdburitype
- Httpuritype
- Timestamp with local time zone
- Timestamp with time zone
- Urowid
- Day to second
- Year to month
- Nclob
- Longvarchar

## Step 1. Create a Data Coverage Task

To analyze the data coverage across combinations of data values, ranges, or groups, create a data coverage task.

1. Log in to Test Data Manager and open the data set that contains the data that you want to analyze.
2. Click the **Data Coverage** tab.

3. Click **Actions** > **New**.

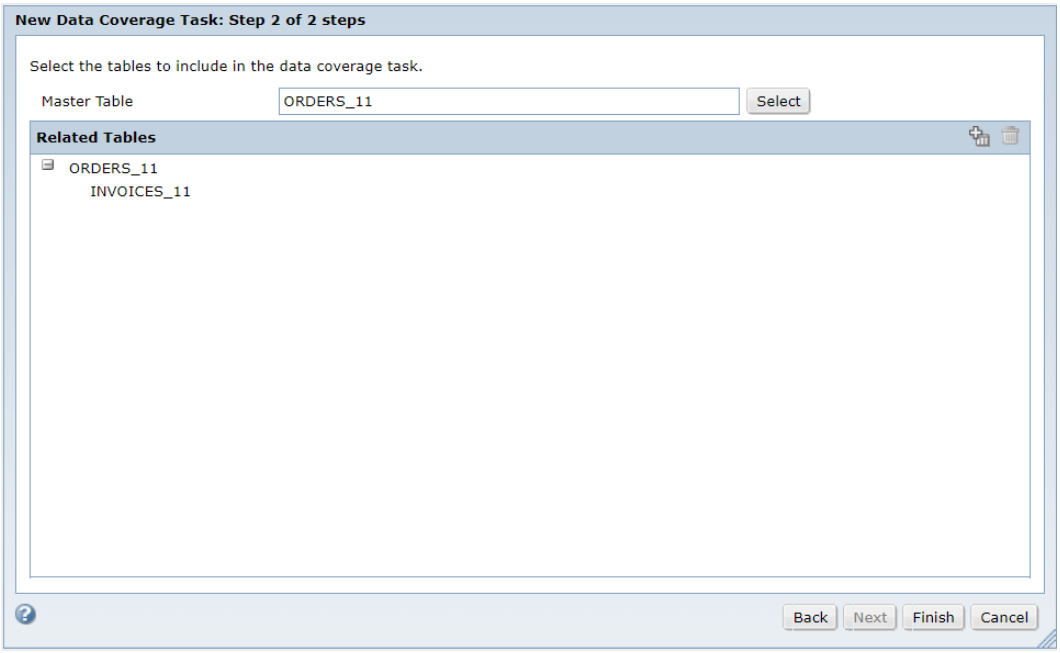
   The **New Data Coverage Task** window appears.

4. Enter a name and optional description for the data coverage task.
5. Enter the minimum threshold value.

   The minimum threshold represents the minimum number of values you require to consider the data density adequate.

6. Click **Next**.
7. Select the master table for the data coverage task and click **OK**.

   The related tables appear in the **Related Tables** list. You can remove tables that you do not want to include in the analysis.
   The following image shows the second step of the data coverage task wizard:




8. Optionally, you can select a related table and click the **Add Tables** button to add tables related to the table.

   You can enter an alias name if you want to use a different name to identify the table in the task. Alias names must be unique within a task. If you use a single table multiple times, use a different alias each time.

9. Click **Finish** to create the task.

   The data coverage task page opens with a blank graph.

You can now configure the data coverage task. Create the columns that you want to plot on the graph to perform the analysis.

## Step 2. Create Columns in the Data Coverage Task

Create the columns that you require to perform the analysis. To perform the analysis required in the example, create a column using ranges and a column using the data as is. You can then analyze the data available for specific ranges of date values in the data.

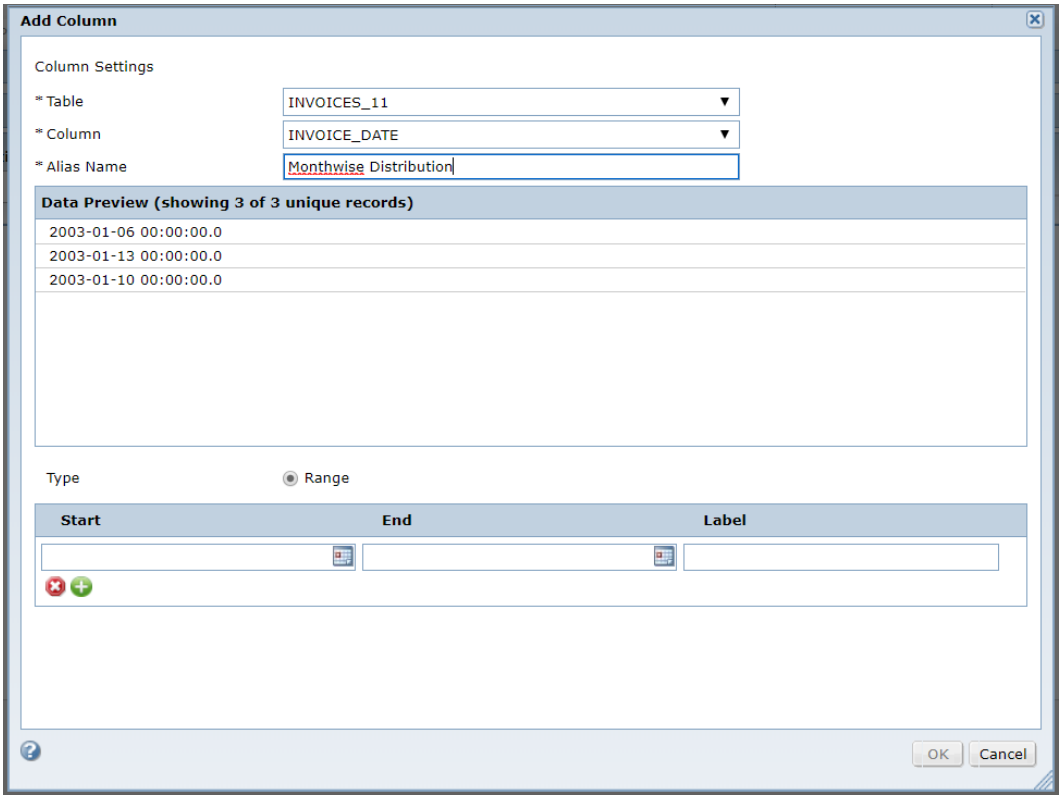
1. To create a column that you want to plot on the graph, click the **Add** button on the **Columns** pane.

   The **Add Column** window opens.

2. From the list of tables, select the table that contains the column.
3. From the list of columns, select the source data column to which this column refers.

   The column data appears in the **Data Preview** pane.
4. Enter an alias name for the column.

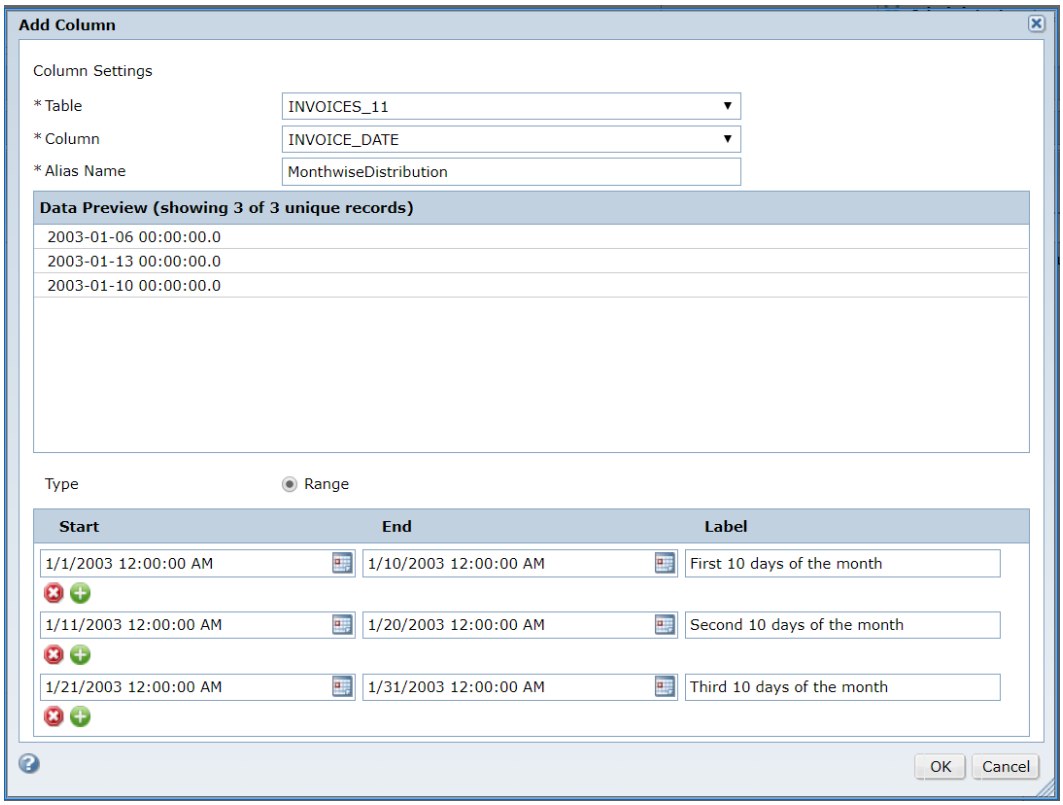
   Column aliases must be unique within a task.
   The following image shows the **Add Column** dialog box:




5. Select the **Range** type.
6. To create a range, enter the start value of the range in the **Start** field and the end value of the range in the **End** field.

   The start value must be less than the end value. Values in different ranges cannot overlap.
7. Enter a label for the range in the **Label** field.
8. Click the **Add** button to add another range.
9. Repeat steps 6 through 8 to create the required number of ranges.

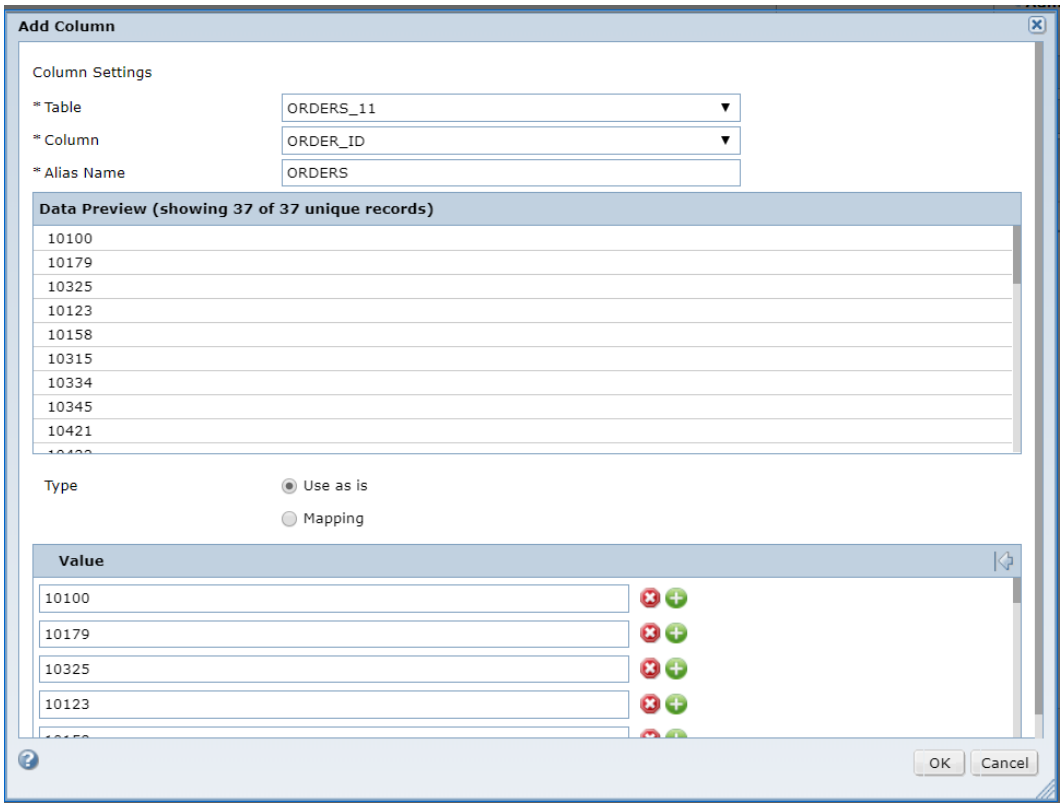
The following image shows the **Add Column** dialog box with the required ranges:




10. Repeat steps 1 through 4 to create another column.
11. Select the **Use as is** type.

    The data from the preview pane is added to the **Value** tab.
12. Optionally, you can edit the data in the **Value** tab. Click the **Reset** button to replace the edited data with data from the preview pane. Click the **Add** or **Remove** button to add or delete individual values.

The following image shows the **Add Column** dialog box with data values:



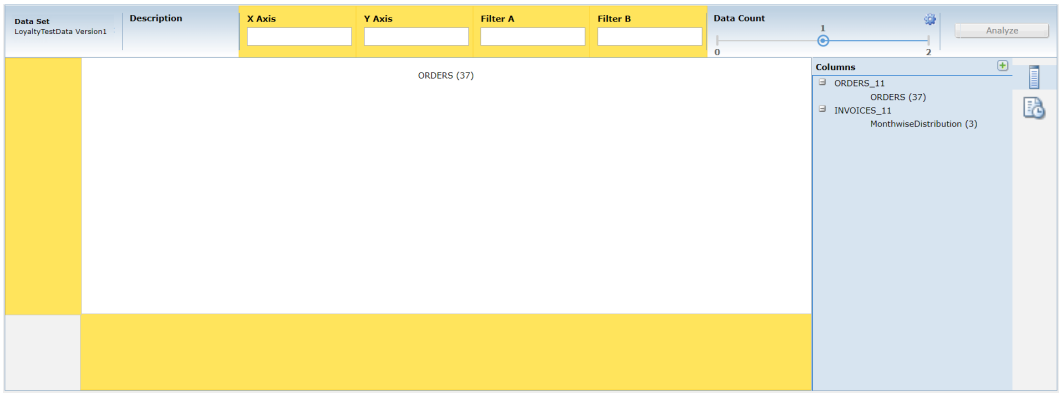

13. Click **OK**.

## Step 3. Perform the Analysis

Use the columns that you create and the settings in the data coverage task to analyze the data coverage.

1. On the **Columns** pane, select a column that you want to add to the X axis on the graph and click **Move to X**. You can also click the required column name from the list of columns and drag the column into the **X Axis** field.

   The following image shows the data coverage task page with the columns created:



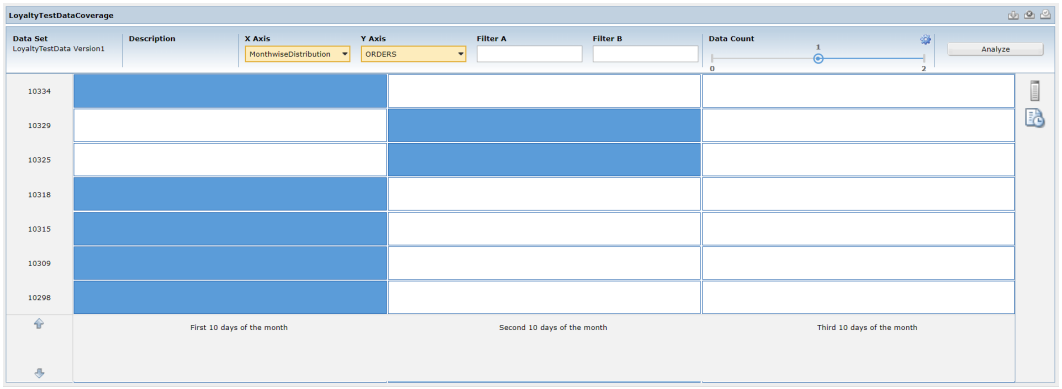

   The column values are plotted along the X axis.

2. Select and add a column to the Y axis.

The column values are plotted along the Y axis.

3. Click **Analyze**.

   The data coverage analysis results appear on the graph. Data coverage is represented based on the minimum and maximum data count values that you set. The color of the cell indicates the density of data. White indicates less than required data. Light blue indicates data within the required data range. Dark blue indicates a higher data density than the minimum threshold that you set.
   The following image shows the data coverage analysis results page:




4. Optionally, you can click the **X Axis** field or the **Y Axis** field and select specific cells from the list to view specific results.

## Step 4. Update the Data

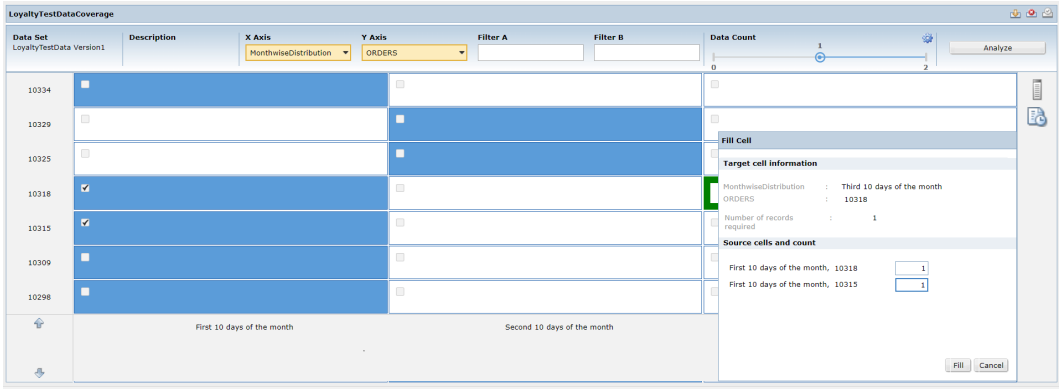
Update data across cells to add data where you need it or to move excess data. When you edit data across cells in the data coverage task, the data is updated in the data set.

1. On the graph, select the cell to which you want to add data. The cell is highlighted with a green border.
2. Right-click the selected cell and click **Fill**.

   The **Fill Cell** dialog box opens.

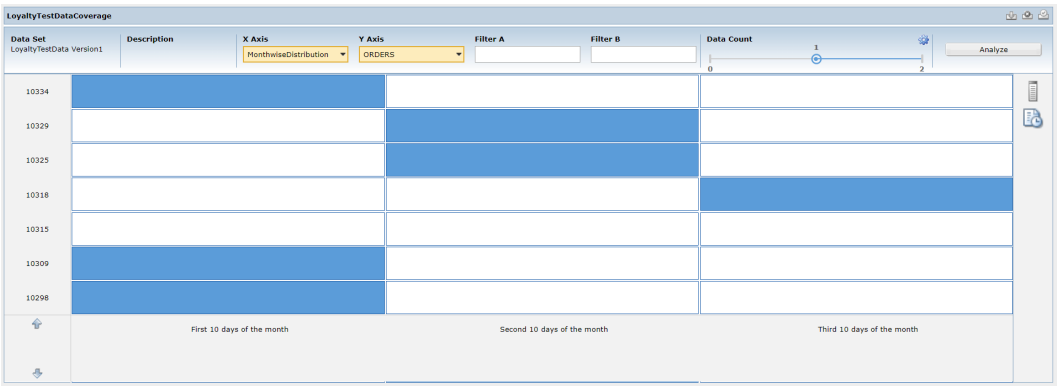
3. Select the source cell from which you want to move data on the graph. You can select multiple source cells.

   The **Fill Cell** dialog box lists the source cells that you select and suggests the number of rows to move. You can enter a different number.
   The following image shows the **Fill Cell** dialog box on the data coverage analysis page:




4. Enter the number of rows that you want to move from each source cell.
5. Click **Fill**.

A separate job runs to fill data from each of the source cells.
The following image shows the data coverage analysis page after the data update task completes:




6. If the data contains fields that need input, a dialog box opens with fields to enter the data. Enter the data and click **OK**.

   The data from the source cells is edited to fill the selected cell. If you close the task before the user input dialog box appears, the job pauses. When you open the task, you must click the **User Input** button to view and enter the required data.

You can verify that the data is updated correctly after the task completes. Click the **Analyze** button without making any changes to the task. The results plotted on the graph must not change.

# Author

**Sadhana Kamath**